



Review Article

Kynurenic acid derivatives in treatment for Alzheimer's disease

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ABSTRACT

This review provides a comprehensive overview of kynurenic acid (KYNA) derivatives with specific focus on KYNA amides as an emerging class of multi-target directing drugs (MTDDs) for the treatment of Alzheimer's disease (AD). It highlights the urgent need for novel AD therapies, discusses the key structural and pharmacological attributes that position KYNA as a promising candidate for this role, outlines laboratory synthesis for KYNA scaffold. Furthermore, the review summarizes the most promising KYNA derivatives reported in the literature, critically evaluates the results from conducted bioassays, and establishes future perspectives for KYNA MTDDs. Overall, eleven leading structures (compounds 1–11) were identified whose MTDD profiles usually combine the neuroprotective and anti-inflammatory features of KYNA with neurotransmitter modulation, anti-amyloid-beta ($A\beta$) aggregation activity, and, less frequently, antioxidant properties and the maintenance of ion homeostasis. Comparative analysis of the bioassay data identifies compounds 2 and 6 as MTDDs with superior translational validity, evidenced by their demonstrated efficacy *in vivo* models (*Caenorhabditis elegans*). Conversely, compounds 7 and 8 emerged as the candidates with the highest potency and broadest pharmacological scope. These derivatives effectively modulated five distinct pathological hallmarks of AD: $A\beta$ accumulation, oxidative stress, neurotransmitter imbalance, and neuroinflammation. Notably, they exhibit disease-modifying potential by functioning not only as neuroprotective agents but also as promoters of neurogenesis. Synthetically, amidation represents the predominant strategy for achieving an MTDD profile, facilitating the efficient modification of pharmacological activity in a single step *via* the incorporation of bioactive amines.

1. Theoretical background

To provide immediate context, we first summarize the theoretical background underlying this review, specifically: the limitations of conventional monotherapies, the rationale for the multi-target-directed drugs (MTDDs) approach, and the chemical and biological advantages of kynurenic acid (KYNA) for MTDD application. For a more granular examination of these mechanisms, readers are referred to the subsequent detailed discussions in chapters Current state of treatments for Alzheimer's disease and Derivatives of kynurenic acid as potential multi-target directing drug.

1.1. The summary from theoretical background

1.1.1. The Limitations of monotherapy in Alzheimer's disease

Alzheimer's disease (AD) is characterized by a multifactorial etiology involving complex, interconnected pathogenic cascades *e.g.* amyloid-

beta ($A\beta$) aggregation, tau hyperphosphorylation, oxidative stress, neuroinflammation, and neurotransmitter imbalance. Despite extensive research, therapies targeting isolated pathological markers, such as cholinesterase inhibitors and NMDA antagonist for the restoration of neurotransmitter balance or monoclonal antibodies for $A\beta$ clearance, have demonstrated limited clinical efficacy. This failure is largely attributed to the reductionist approach of monotherapy, which neglects the synergistic nature of AD pathology. Eliminating a single pathologic feature is insufficient to arrest the downstream neurodegenerative processes.

1.1.2. The rationale for multi-target-directed drugs

To address the complexity of AD pathology, drug discovery has shifted toward MTDDs. Unlike combination regimens (polypharmacy), which are associated with varying pharmacokinetic profiles and an increased risk of adverse drug interactions, MTDDs are single chemical entities designed to simultaneously modulate multiple distinct targets

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within the neurodegenerative network. This polypharmacological approach aims to restore systemic homeostasis by acting synergistically on key pathological nodes thereby offering a superior therapeutic profile compared to agents with high selectivity for a single biological target.

1.1.3. Kynurenic acid: a privileged yet challenged scaffold

Kynurenic acid (KYNA) represents an attractive scaffold for MTDD discovery, combining biological relevance with exceptional chemical tractability. Structurally, KYNA possesses a privileged quinolin-4-one core (tautomeric with 4-hydroxyquinoline) that allows for extensive structural diversification via robust synthetic protocols, such as the Conrad-Limpach reaction. Biologically, it functions as an endogenous neuroprotectant, counteracting excitotoxic quinolinic acid (QUIN) through NMDA receptor antagonism and $\alpha 7$ -nicotinic receptor modulation and it is key mediator of inflammation regulation. However, the clinical application of unmodified KYNA is constrained by the “kynurenic acid paradox,” where therapeutic concentrations risk inducing cognitive deficits via global receptor blockade. Additionally, native KYNA suffers from poor pharmacokinetic profiles, including negligible BBB permeability and slow receptor dissociation kinetics. Therefore, leveraging its accessible synthetic chemistry to generate optimized derivatives offers a strategy to retain neuroprotective efficacy while overcoming bioavailability barriers and the limited therapeutic window.

1.2. Current state of treatments for Alzheimer's disease

AD is the most common form of dementia, causing progressive cognitive decline in memory, reasoning, language, behaviour, personality, and spatial understanding. AD accounts for 60–80% of all dementias, and due to an increased average lifespan, the risk of AD increases as well, making it one of the greatest challenges to modern medicine.¹ Despite enormous effort in the past decades to develop a causal treatment, this milestone has yet to be achieved. It has proven to be difficult to find an effective treatment due to complex pathology of AD (Fig. 1), which incorporates hallmark protein accumulations, specifically extracellular amyloid-beta ($A\beta$) plaques and intracellular neurofibrillary tangles (INT) formed by hyperphosphorylated tau protein. Other contributors are neurological inflammation, oxidative stress, infectious diseases, changes in the gut microbiome, synaptic dysfunction and imbalance of neurotransmitters such as acetylcholine, noradrenaline, serotonin, and glutamate.²

Based on these pathological contributors to AD several etiological hypotheses have been developed over the years. The oldest one is the cholinergic/glutamatergic hypothesis, which states that the loss of neurons is proportional to the decline of neurotransmitters, especially in basal forebrain area where great number of cholinergic interneurons are located, and that overstimulation of the glutamatergic NMDA receptor causes excitotoxicity.³ Most of the initially approved drugs for AD revolve around this hypothesis. The amyloid hypothesis states that the

aggregation of $A\beta$ peptides (most common are $A\beta 38$, $A\beta 40$ and $A\beta 42$) is the primary cause of neurodegeneration in AD.⁴ However, this hypothesis was challenged with the introduction of anti- $A\beta$ monoclonal antibodies. Despite their efficacy in removing plaques, their ability to slow down neurodegeneration in AD was shown to be moderate at best.^{5–7} Furthermore, this hypothesis was overshadowed by the retraction of a frequently cited article in Nature, that claimed to identify a specific $A\beta$ type ($A\beta^{*56}$) responsible for the memory loss. These claims were found fraudulent.⁸ Currently $A\beta$ plaques are considered as suitable environmental factors for the neurodegeneration rather than the etiological cause itself. This shifted the research focus in recent years to the second hallmark protein accumulation, specifically hyperphosphorylated tau protein.⁹ Numerous strategies can be employed to remove INT¹⁰ such as.

- tau immunotherapy (monoclonal antibodies and vaccination).
- antisense oligonucleotides (they impair function of tau mRNA).
- Aggregation inhibitors (they bind to tau monomers or oligomers preventing them from sticking together).
- kinase inhibitors (they prevent hyperphosphorylation of tau protein).
- microtubule stabilizers (they substitute the role of hyperphosphorylated tau protein) There are other, less developed hypotheses or treatment approaches e.g. neuroprotection by affection of regenerative mechanisms of human body.^{11,12} The remainder of hypotheses revolve around the rest of the known pathological contributors to AD, namely neuroinflammation,¹³ oxidative stress,¹⁴ ion homeostasis¹⁵ etc.

All the commercially available AD drugs¹⁶ target only one pathological aspect of AD and can be generally divided into two groups. The first are symptomatic drugs such as Donepezil, Rivastigmine, Galantamine, and Memantine (Fig. 2). This group modulates neurotransmitter function. Specifically, Memantine, as a non-competitive antagonist of the NMDA receptor with a binding site deep in the open ion channel, regulates glutamate activity, preventing excitotoxicity and mitigating neuronal damage. Its binding is reversible with fast on/off rates, which is crucial for its therapeutic profile. It can reduce agitation and hallucinations in moderate to severe AD. All the other neurotransmitter modulators from this group are cholinesterase inhibitors. They prevent acetylcholine breakdown, enhancing cholinergic neurotransmission. This temporarily improves memory, attention, and daily function, typically for six to twelve months.

The second group are causal drugs such as Aducanumab, Lecanemab, and Donanemab. These disease-modifying drugs focus on the amyloid hypothesis, specifically, they are monoclonal antibodies, that target $A\beta$ plaques. However, their efficacy was shown to be only moderate.

Most of the drugs currently in the clinical trials (Fig. 3) focus on other potentially causal contributors to AD; the most frequent are anti-

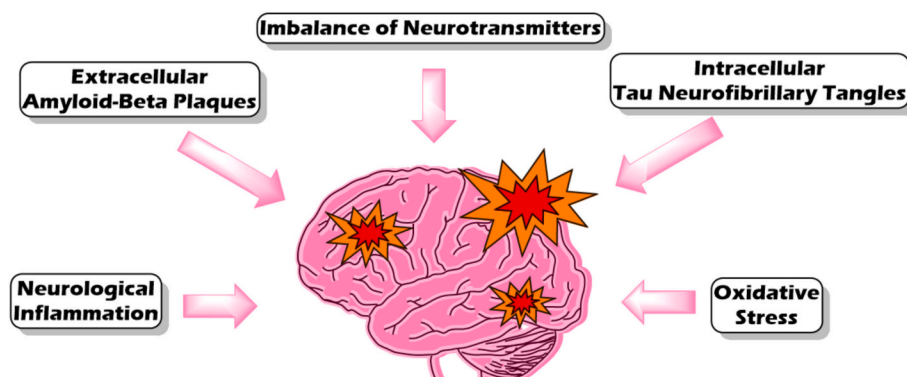


Fig. 1. Main pathological contributors to the progression of Alzheimer's disease.

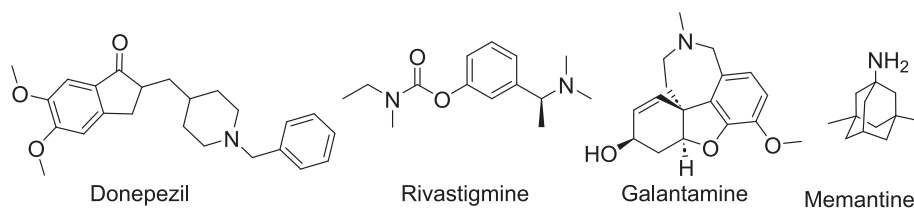


Fig. 2. Structures of symptomatic drugs approved by FDA for AD.

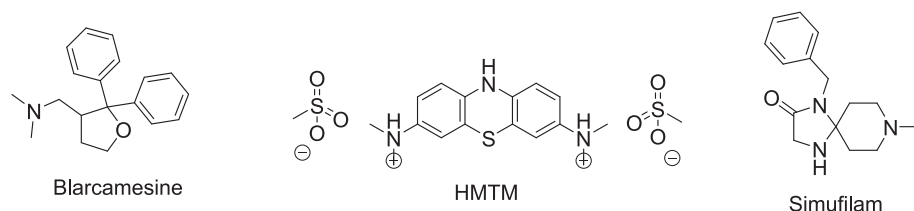


Fig. 3. Structures of experimental drugs currently in clinical trials for AD.

tauopathies. Specifically, Hydromethylthionine mesylate (HMTM)¹⁷ belongs to the aggregation inhibitors group, Semorinemab¹⁸ and Gosuranemab¹⁹ represent tau immunotherapy approach and BIIB080²⁰ from company Biogen is antisense oligonucleotide, which switches off tau production inside the nerve cells. However, the results from the clinical trials about their efficacy are mixed.

Sargramostim²¹ is a recombinant granulocyte macrophage colony-stimulating factor (GM-CSF). It was originally approved for the stimulation of the white cells production in patients undergoing chemotherapy and now it is being repurposed for AD as a modulator of the immune system to prevent neuroinflammation. Blarcamesine²² is a sigma-1 receptor (S1R) agonist. Stimulation of S1R triggers a cascade of beneficial neuroprotective effects and restores cellular homeostasis. Simufilam's²³ proposed mechanism of action is to bind to the structural

protein Filamin A (FLNA) in neurons, which is altered by AD progression. This subsequently reduces downstream neurodegeneration; however, a concern was raised about possible image and data manipulation in preclinical papers and FDA Citizen Petition (FDA-2021-P-0930) was filed. Even though this petition was denied by FDA and Simufilam is currently in phase III of clinical trials, it helped to bring public and scientific scrutiny to a possible issue. Thus far all these experimental drugs were successful in phase II of clinical trials, their efficacy is yet to be determined in phase III.

The complexity of AD, combined with the low to mediocre efficacy of currently FDA approved drugs and the mixed results in clinical trials from experimental drug, highlights the potential for the future utilization of MTDDs in the treatment of AD. Rather than relying on drug cocktails with their inherent risks of toxicity and mismatched

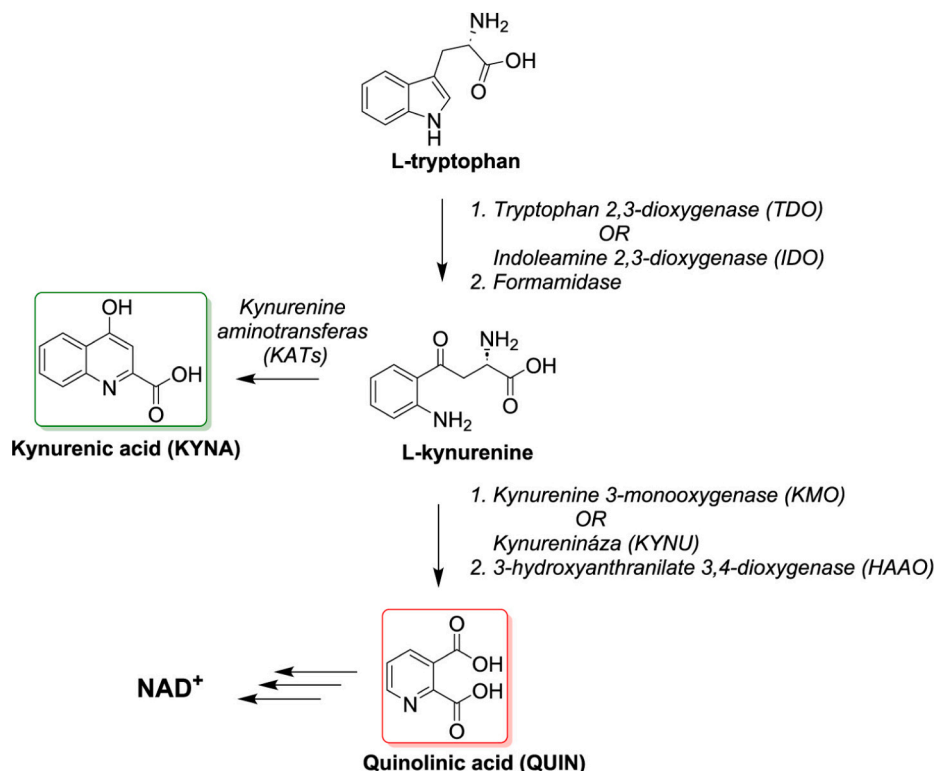


Fig. 4. Selected part of kynurenine pathway.

bioavailability, MTDDs integrate multiple mechanisms of action into a unitary chemical scaffold designed to modulate the complex neurodegenerative network.

1.3. Derivatives of kynurenic acid as potential multi-target directing drug

1.3.1. Endogenic kynurenic acid

Kynurenic acid (KYNA) is a key metabolite of kynurenine pathway (KP) in the human body. KP was initially considered only as the main metabolic pathway for L-tryptophan (TRP) to generate the redox cofactor nicotinamide adenine dinucleotide (NAD⁺) (Fig. 4) and KYNA was dismissed as a biologically inert by-product of TRP oxidation following its discovery in 1853 by Justus von Liebig. Only recently KP was established as a fundamental interface between the central nervous system (CNS) and the systemic environment, particularly the immune system. Its activity is induced by physiological stressors, most notably inflammation, making it a key mediator of the brain's response to peripheral insults and a central component of the gut-brain-immune axis. KYNA was recognized as a potent neuroprotective agent and as an antagonist of the NMDA receptor, similar to Memantine. Specifically, KYNA is competitive antagonist at both glutamate and strychnine-insensitive co-agonist glycine binding sites,²⁴ however the reversibility of the latter is slow enough to be considered non-competitive.²⁵ Another key metabolite of KP, specifically quinolinic acid (QUIN) is considered to be neurotoxic, and it is an agonist of the NMDA receptor. Since KYNA and QUIN have opposite biological effects, their relative concentrations are also called the neuroprotective ratio, which maintains brain homeostasis and can be disrupted in pathological conditions. The role of other metabolites from KP, individual enzymes and the intricate interconnections of KP to biochemistry of human body even in relation to AD were already discussed in great detail in other reviews.^{25–28}

There are indications that overproduction of KYNA is one of the immune responses to slower clinical progression in AD as the elevated concentration was found in many AD patients.²⁹ Furthermore, inhibitors of kynurenine 3-monooxygenase (KMO) have been shown to have positive effect in AD and Huntington disease.³⁰ Inhibition of KMO blocks production of downstream metabolites in KP and indirectly increases KYNA concentrations. Metabolites of KP, predominantly KYNA can be utilized as biomarkers for more efficient detection of AD in early stages.³¹

1.3.2. The kynurenic acid paradox

Despite all above-mentioned positive arguments for the KYNA as potential future lead structure against AD recent evidence suggests this approach is not without risk. The elevated KYNA blocks excitotoxic cell death, supraphysiological levels can paradoxically impair learning and memory by excessively inhibiting NMDA and $\alpha 7$ -nicotinic receptors.³² This trade-off between neuroprotection and cognitive function creates a narrow therapeutic window. Modern drugs might be more focused on restoration of the homeostatic QUIN/KYNA ratio without inducing the global receptor blockade associated with cognitive deficits.

Furthermore, unmodified KYNA has certain inherent disadvantages, especially if it is to be used as a drug. Its solubility in water is low,³³ it crosses the blood-brain barrier (BBB) very poorly (approx. 0.2×10^{-6} cm/s determined by *in situ* brain perfusion method)³⁴ and the reversibility of its antagonistic binding to NMDA receptor is very slow compared to Memantine, which could cause already mentioned chronic inhibition of glutamatergic neurotransmission. All these undesirable properties can be improved by rational molecular design and synthesis of KYNA derivatives. Ultimately KYNA derivatives are uniquely positioned as a privileged scaffold for MTDDs design because their core structure is an endogenous metabolite that naturally interfaces with multiple AD-associated targets and rational synthetic design ushers additional bioactivity and modification of pharmacokinetic properties.

2. Review

2.1. The laboratory synthesis of kynurenic acid and its derivatives

Kynurenic acid can exist in two tautomeric forms: the oxo form, 4-quinolone-2-carboxylic acid, and the enol form, 4-hydroxyquinoline-2-carboxylic acid. The broader structural motif of KYNA is the quinolin-4-one, which is privileged heterocyclic pharmacophore with well-established utilization in medicinal chemistry.³⁵ There are at least nine robust synthetic approaches to obtain this heterocycle.³⁶ However, only a few of them are established in the literature to directly yield quinolin-4-one with a carbonyl group in position 2 in a consistent and high-yielding manner (Fig. 5). Specifically, these are Conrad–Limpach synthesis and the Biere–Seelen synthesis. The remaining synthetic approaches³⁶ (e.g. Gould–Jacobs reaction, Snieckus synthesis, Camps cyclization, Pd-catalyzed carbonylation of 2-iodoaniline and terminal acetylenes, or *N*-heterocyclic carbene-catalyzed dehydrative acetylation of 2-aminobenzaldehyde with α -halogenketone and subsequent intramolecular heterocyclization) typically yield quinolin-4-one with an alkyl group in position 2. This side-chain alkyl group must be transformed into a carboxylic acid in subsequent oxidation steps.

2.2. Retrosynthetic logic

A SciFinder® database search reveals that the most frequently used of the three methods is the Conrad–Limpach synthesis, utilizing dialkyl acetylenedicarboxylate **V** and an aniline derivative **III** as substrates. Alternatively, combination of dialkyl 2-oxosuccinate **IV** and an aniline derivative **III** can be used as substrates. Retrosynthetic analysis (Fig. 5) of both combinations involves the cleavage of C3–C4 and N1–C2 bonds tracing the quinolone core **I** back to the acyclic precursor **II** formed by the condensation of the aniline derivative **III** with the four-carbon synthon **IV** or **V**. The limiting step of this synthesis is usually cyclisation (C3–C4 bond formation) which requires high temperature around 250 °C and inert high-boiling solvent (e.g. diphenyl ether, Dowtherm™). These thermal cyclization conditions are unifying procedural feature of the Conrad–Limpach and Gould–Jacobs reaction as both reactions share the identical mechanism of the second cyclisation step. This high cyclization temperature poses the main disadvantage of the method as it may result in carbonization of starting material and lowering the final yield. This drawback may be partially sidestepped by executing the reaction one-pot without the isolation of intermediate **II**.³⁷

Also noteworthy is the possible regioselectivity of this reaction when a meta-substituted aniline **X** is used as the starting material (Fig. 6). Due to the electrons from the benzene ring acting as the nucleophile in the cyclization step, two possible isomers can form. Typically, 7-substituted isomer **XI** is the only product, however, the formation of a regioisomeric mixture was observed for $X = F, Cl, Me, CF_3$, and NO_2 . Additionally, for $X = CN$ the 5-substituted isomer **XII** was the sole product.³⁸

Based on the SciFinder® database search the Biere–Seelen synthesis appears significantly less frequently in the literature and is utilized mostly in patents. This method yields a quinolin-4-one with carbonyl group in position 2 and 3. The Biere–Seelen synthesis is very similar to the Conrad–Limpach synthesis. The key difference in the retrosynthetic analysis (Fig. 5) includes the cleavage of C2–C3 instead of C3–C4. Subsequent cleavage of the N1–C2 bonds traces the quinolone core **VI** back to the acyclic precursor **VII** formed by the condensation of the alkyl anthranilate **VIII** with dialkyl acetylenedicarboxylate **V**. Furthermore, the second cyclization step from **VII** to **VI** is mediated by a strong base, rather than high temperature, and there is no regioselectivity preference based on the starting material.

From a retrosynthetic logic, the third option is the transformation of a functional group, specifically an alkyl to a carboxylic acid (Fig. 5). On the one hand, this method offers greater synthetic variability as starting quinolin-4-ones with an alkyl group at C2 are accessible by a plethora of synthetic approaches. On the other hand, this option inherently adds up

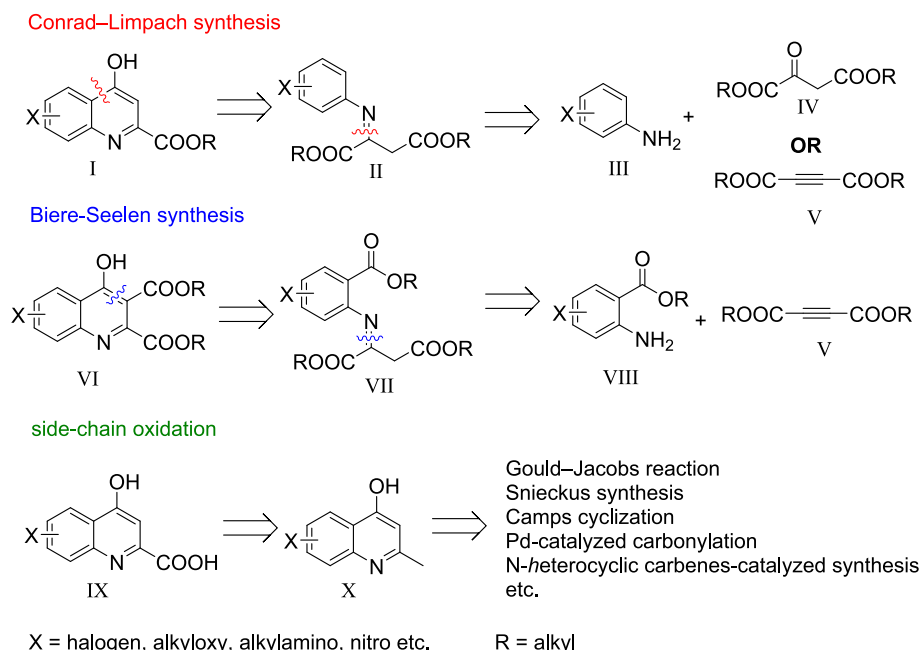


Fig. 5. Disconnection approach to KYNA framework synthesis.

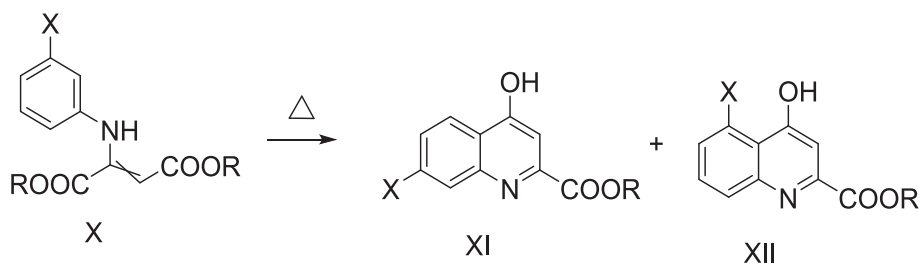


Fig. 6. Regioselectivity of the thermal cyclization step in Conrad–Limpach and Gould–Jacobs reactions.

to two steps to synthesis. These added oxidation steps can be performed either *via* direct oxidation with potassium permanganate, or in two steps with selenium dioxide *via* an aldehyde or by oxidation with chromium trioxide in acetic anhydride with subsequent hydrolysis.

2.3. Examples of concrete synthetic procedures

The retrosynthetic analysis wouldn't be complete without showcasing concrete reaction conditions on individual reaction steps (Fig. 7). Specific procedures, work-ups, and separation techniques can be found as follows: for **XV**,³⁹ **XVIII**,⁴⁰ **XX**,⁴¹ and **XXII**.⁴² Notably, the reference for the Biere-Seelen synthesis remains the original paper from 1976. Based on the SciFinder® search this procedure has been referenced in only couple of journals since then, which might question the practicality of the method. Despite multiple theoretical oxidation conditions for converting **XXI** to form **XXII**, in the literature only a two-steps process with selenium dioxide *via* an aldehyde is described. The reaction conditions for the second oxidation step from aldehyde to carboxylic acid are described in the literature as somewhat cumbersome.⁴² However, for the structurally related 4,7-dihydroxy-1,10-phenanthroline-2,9-dicarbaldehyde more straightforward oxidation conditions with hydrogen peroxide are known.⁴³

2.4. Comparison of KYNA syntheses, green chemistry, and modern alternatives

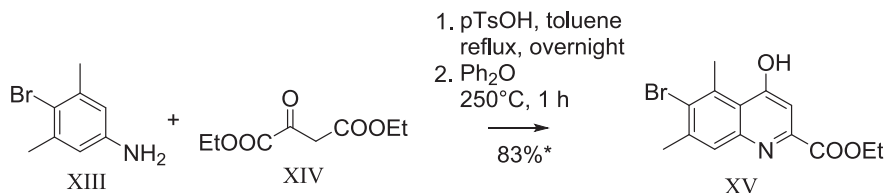
Considering the very limited data on variability and robustness of the

Biere-Seelen synthesis and alkyl side-chain oxidation for quinolin-4-ones, the preferred method is the Conrad–Limpach synthesis. The limiting step and the main hurdle for the scale up of this reaction is the thermal (approx. 250 °C) cyclisation step. Historically used solvents for thermal cyclisation such as diphenyl ether and Dowtherm™, are difficult to remove from the reaction mixture. They require large volumes of non-polar solvent (hexane or petroleum ether) to precipitate the product during the work-up which makes them non-green solvents. To lower the reaction temperature (approx. 50 °C) and increase yield, acidic conditions using Eaton's reagent or PPA (polyphosphoric acid) can be used,⁴⁴ however, they are viscous, corrosive and create large amounts of waste during the neutralization work-up. To achieve lower reaction temperature (approx. 120 °C) it is also possible to employ copper catalyst in dichloroethane (DCE)⁴⁵ which is halogenated solvent. To circumvent these drawbacks the Conrad–Limpach reaction has been utilized as optimization template for green chemistry considerations. Specifically, a reaction conditions employing microwave irradiation with green solvents (diethyl carbonate and γ -valerolactone) have been developed.⁴⁶ As for the scale up, it is possible to conduct the reaction in the flow system (300–360 °C, 100–160 bar, 0.45–4.5 min) with THF as low-boiling solvent with an average yield of 74%.⁴⁷

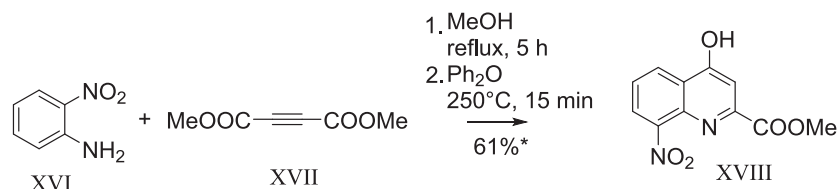
2.5. Current state of kynurenic acid derivatives in treatment for Alzheimer's disease

The literature search was conducted using SciFinder® database, specifically by substructure search of KYNA molecule with keywords

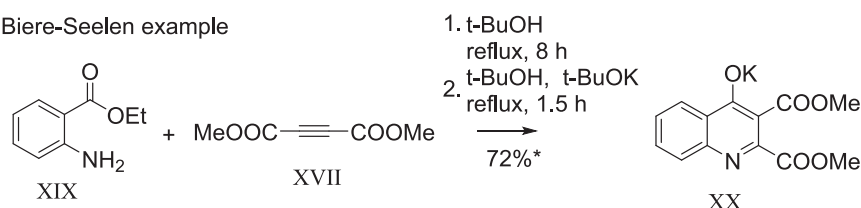
Conrad–Limpach with dialkyl 2-oxosuccinate example



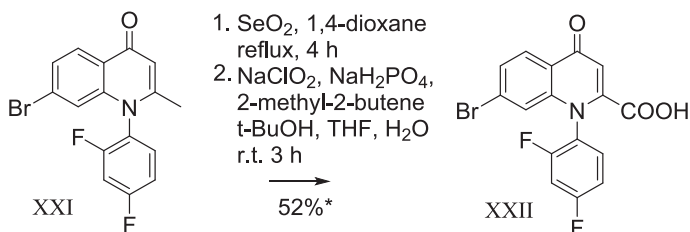
Conrad–Limpach with dialkyl acetylenedicarboxylate example



Biere-Seelen example



alkyl side-chain oxidation example



*cumulative yield after the second step

Fig. 7. Synthesis of KYNA and its derivatives, examples from literature.

“Alzheimer”, “dementia”, and “brain”. As this setup returns a great number of hits (around 2153 journals and 242 patents) it was refined to include only journals with substructure of KYNA amide (30 journals and 79 patents) and most of them are discussed in this chapter. Although the KYNA scaffold is amenable to various chemical modifications, such as substitution at the C3 position^{48,49} or the incorporation of an altered heterocyclic core.⁵⁰ KYNA amides were selected as the primary focus. They represent one of the largest groups of KYNA derivatives and are

uniquely positioned to achieve a multi-target profile in a single step via the incorporation of bioactive amines. Based on a literature survey, we can classify the most frequent modifications of the KYNA scaffold on three distinct centers (Fig. 8). Either in positions 5–8 on quinoline ring by nucleophilic aromatic substitution or by utilizing already substituted starting materials (derivatization A). Similar substitution of the hydroxy group can be conducted on the carbon in position 4 in a few of steps (derivatization B). The third modification is amidation of the carboxylic

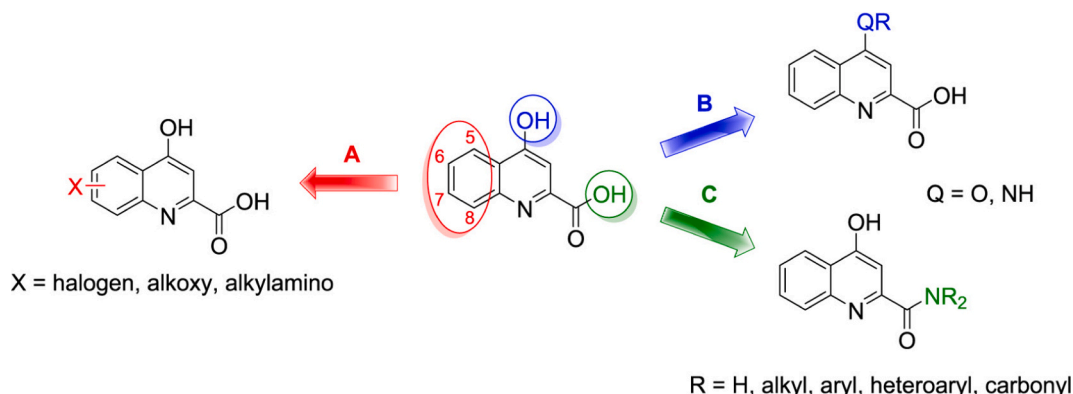


Fig. 8. Depiction of the most frequent chemical modification to KYNA molecule.

acid moiety (derivatization C).

KYNA derivatives that were reported in the literature to possess the most promising attributes to be MTDDs as they achieved the best results in various bioassays are depicted in Fig. 9. All of the reported derivatives were either prepared by Conrad-Limpach reaction, usually from substituted anilines to achieve derivatization A on quinoline ring or various KYNA isomers were purchased from supplier.

The first three compounds 1–3 (Fig. 10)51 are the most promising candidates from a broader series of KYNA derivatives (derivatization A) that includes KYNA amides. The conducted bioassays included their binding ability to NMDA and mGluR5 receptors, acetylcholinesterase (AChE) inhibition, 2,2-diphenyl-1-picrylhydrazyl (DPPH) radical scavenging to assess their ability to prevent oxidative stress. Their ability to interfere with the A β fibrillation process was tested as well, along with protection against A β -induced toxicity in transgenic roundworms (*Caenorhabditis elegans*) which express full-length A β ₄₂. Lastly, BBB permeability was assessed by MDR1-MDCKII cell-based model. Compound 1 is the best AChE inhibitor from the series with an IC₅₀ of 10.9 μ M; however, its BBB permeability is low. Specifically, the BBB permeability is similar to naturally occurring antioxidant Epigallocatechin gallate (EGCG) that was used as reference compound. Both 2 and 3 showed high BBB permeability (approx. 400-fold that of EGCG), 2 displayed greater antioxidant potential than EGCG and exhibited 57.1% reduction in A β ₄₂ fibrils. On the other hand, 3 has only moderate antioxidant potential compared to EGCG, but good results in antagonistic binding to NMDA (42.4% inhibition with a Ki value of 7.98 μ M) and mGluR5 (Ki value 0.829 μ M) receptors. Based on the conducted bioassays, compounds 1–3 as MTDDs focuses on three main pathological contributors to AD, specifically A β accumulations, OS, and imbalance of neurotransmitters, which includes excitotoxicity.

Compounds 4 and 5 (Fig. 11)52 are KYNA amides with a methyl group in position 3. Both were tested as inhibitors of μ -calpain. Calpains are a distinct family of intracellular, Ca²⁺-dependent cysteine proteases that are involved in a variety of calcium-regulated cellular processes, such as signal transduction, cell proliferation, differentiation, and apoptosis. In the case of AD, they are pathologically overactivated, which leads to excitotoxicity.53 Furthermore, excessive activation of

calpains have been implicated as a major component in the signaling cascade that leads to A β production and tau hyperphosphorylation.54 Both achieved the best results from the series with IC₅₀ values of 0.71 $\pm$ 0.07 and 0.73 $\pm$ 0.23 μ M for the μ -calpain inhibition. Compounds 4 and 5 as MTDDs aims to tackle two main potential causes of AD i.e. A β accumulations and neuronal death by excitotoxicity which by extensions partially leads to neurotransmitter imbalance.

Compound 6 (Fig. 12)55,56 is a foldamer that can interfere with the A β aggregation process on multiple stages. The compound consists of four KYNA derivative units. The monomer unit was methyl 4-hydroxy-8-nitroquinoline-2-carboxylate, which underwent derivatization B on the 4-hydroxy group and reduction of the 8-nitro group to an amino group for further derivatization C. Conducted biophysical assays including amyloid kinetics, dot blot, ELISA, and TEM show that 6 effectively inhibits both A β oligomerization and fibrillation. The initial study55 was expanded in 2025 56 to validate its drug-like properties in living organisms, specifically 6 successfully ameliorated symptoms of AD in *C. elegans* models e.g. it reduced paralysis and prevented the loss of sensory function. It also lowered the levels of OS in said roundworms. Using mouse models, 6 is quickly absorbed in the blood plasma and efficiently crosses the BBB. Metabolic stability bioassays showed that 6 is highly stable in human liver microsomes over 24 h. Furthermore, 6 potently inhibits Zn²⁺ mediated A β ₄₂ aggregation by potentially displacing Zn²⁺ thus maintaining ion homeostasis. Conducted bioassays imply that 6 as MTDD targets three relevant aspects of AD i.e. A β accumulations, OS and it showcased neuroprotection in *C. elegans* assay.

Compounds 7⁵⁷ and 8⁵⁸ (Fig. 13) are analogous to KYNA amides and they turned out to be the most efficient from the series that tested not only quinoline but also chromone derivatives. The amide moiety is *N,N*-dibenzyl(*N*-methyl)amine (DBMA) (in 7) which is an active part of experimental drug AP2238. An extensive SAR and computational studies59 were conducted on this drug (AP2238) that evaluated its AChE and butyrylcholinesterase (BuChE) inhibitory activities. In 8, the amide moiety is 2-(1-benzylpiperidin-4-yl)ethan-1-amine, which mimics part of Donepezil structure. Both 7 and 8 were tested as inhibitors of human cholinesterases (hAChE/hBuChE) and β -secretase (hBACE-1), which causes enzymatic cleavage of amyloid precursor protein (APP) in such a manner that pathological A β peptides are formed. Monoamine oxidases (hMAO-A/B) inhibition is important due to the overexpression of these enzymes in AD patients causes oxidative stress and mitochondrial damage. Lipoxigenase-5 (hLOX-5) inhibition was shown to reduce neuroinflammation. The CNS permeability was assessed according to the *in vitro* PAMPA-BBB assay. The last bioassay measured affinity to sigma receptors (σ 1R/ σ 2R) which are associated with neuroprotective and neurogenic brain functions. Both compounds turned out to be the most promising in their respective series. The results from bioassays for 7 were σ 1R (Ki 0.53 $\pm$ 0.07 μ M), σ 2R (Ki 1.30 $\pm$ 0.35 μ M), hLOX-5 (IC₅₀ 30.4 $\pm$ 1.6 μ M), hMAO-A/B (both IC₅₀ above 100 μ M), hAChE (IC₅₀ 4.5 $\pm$ 0.8 μ M), hBACE-1 (IC₅₀ 6.7 $\pm$ 0.8 μ M). For compound 8 σ 1R (Ki 37.4 $\pm$ 6.3 nM), σ 2R (Ki 239 $\pm$ 36 nM), hLOX-5 (IC₅₀ 74.3 $\pm$ 3.2 μ M), hMAO-A (IC₅₀ 15.3 $\pm$ 1.3 μ M), hMAO-B (IC₅₀ 5.2 $\pm$ 0.3 μ M), hAChE (IC₅₀ 46 $\pm$ 4 nM). Furthermore, both are CNS-permeable and in a primary culture of neural stem-cells from the subgranular zone (SGZ) of adult rats they stimulated the differentiation of neural stem-cells to a neuronal phenotype and thus, 7 and 8 could be a promising therapeutic agents promoting brain auto-repair processes. It is also worth mentioning that the corresponding quinoline-based analogues tested inferior in nearly every bioassay, with the exception of hAChE inhibition where they displayed comparable potency. This suggests that the bioisosteric replacement of the scaffold nitrogen with oxygen in 7 and 8 may be advantageous for this specific multi-target profile, despite the deviation from the classical KYNA structure. Compounds 7 and 8 as MTDDs should combine neuro-repair properties and block the first steps of neurotoxic cascades in AD. Conducted bioassays suggest that 7 and 8 hit five relevant aspects of AD i.e. A β accumulations, OS, neurotransmitters imbalance, neuroinflammation and they are not

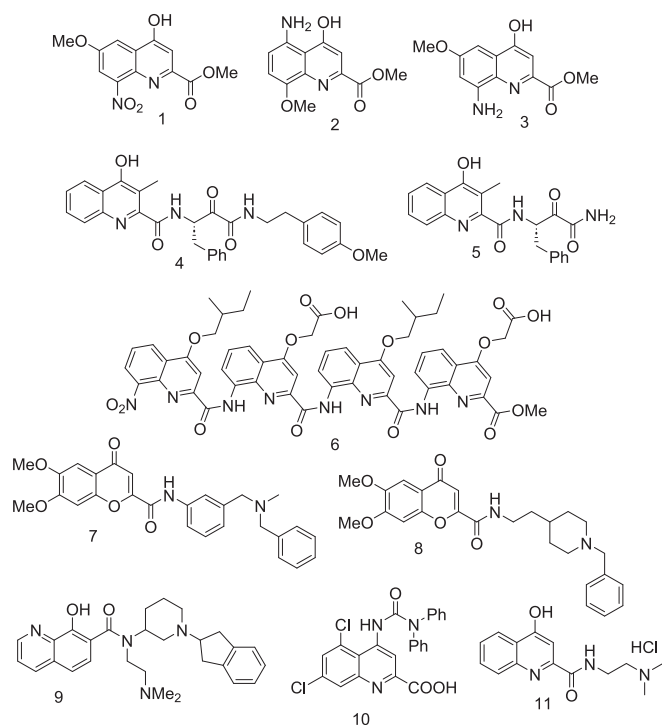


Fig. 9. Structures of the most promising KYNA and KYNA-related derivatives.

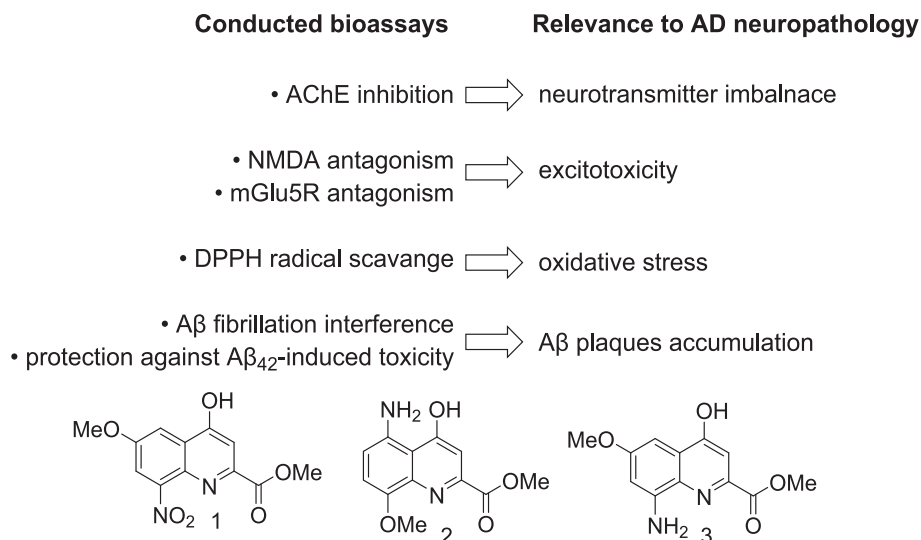


Fig. 10. The lead compounds 1–3 from.51

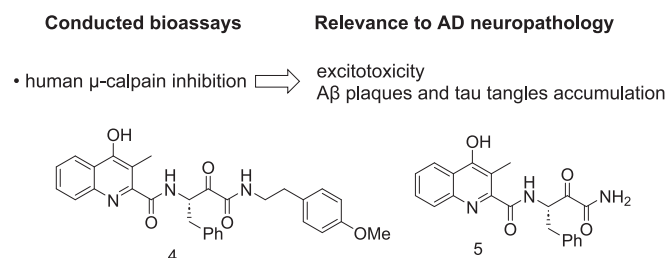


Fig. 11. The lead compounds 4–5 from.52

only neuroprotective, but also stimulate neurogenesis.

Compound **9** (Fig. 14)60 is a regioisomer of a KYNA amide, specifically 8-hydroxyquinoline-7-carboxylate instead of the usual 4-hydroxyquinoline-2-carboxylate. This modification introduces metal-chelating

properties which aim to reduce the redox activity of chelated Cu²⁺ ions and maintain ion homeostasis, which can be disrupted in pathological conditions. Several cations (Zn²⁺, Fe²⁺, Cu²⁺) have been associated with Aβ aggregation. Subsequent hypermetallation of Aβ by these redox-active metals catalyzes the production of hydrogen peroxide leading to significant downstream oxidative stress.61 Furthermore, excessive zinc release in synaptic terminals may play a role in tau hyperphosphorylation.62. **9** was tested for hBuChE and hAChE inhibitory activities with selective affinity and low nanomolar inhibition values (IC₅₀ 11.1 ± 0.6 nM) for hBuChE which is advantageous for the treatment of late AD stages, and reduces side effects (compared to hAChE inhibitors).63 Antioxidant properties of **9** were determined by the DPPH assay with good result (EC₅₀ 544.0 ± 41.5 μM). Furthermore, **9** is not the substrate for the active efflux transport system, as determined in Caco2 cells and it protects neuroblastoma SH-SY5Y cells from toxic Aβ_{1–42} species, which indicates it could be a promising lead as a MTDD.

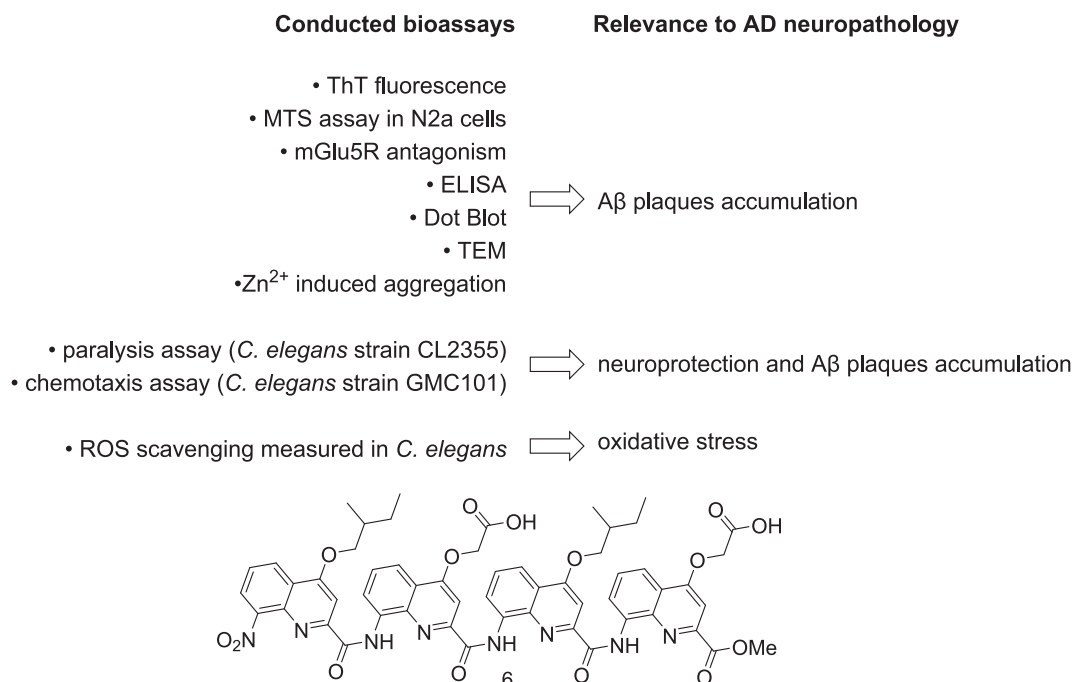


Fig. 12. The lead compound 6 from.55,56

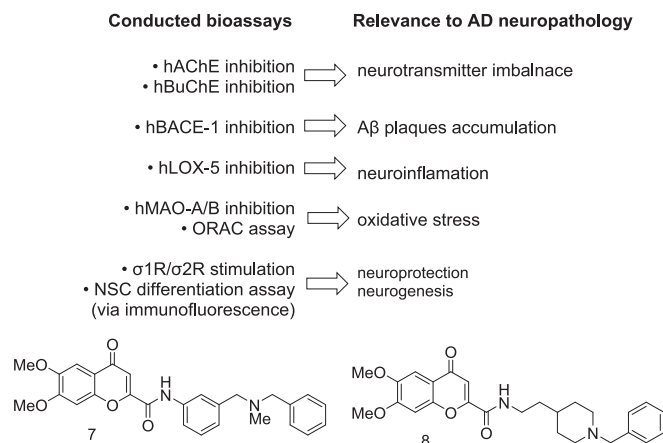


Fig. 13. The lead compounds **7** and **8** from [57,58](#)

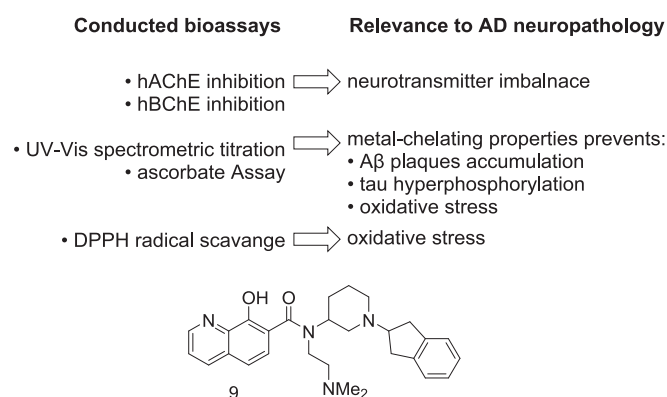


Fig. 14. The lead compound **9** from [60](#)

Notably, the corresponding 2-regioisomer (8-hydroxyquinoline-2-carboxylate) from this study exhibited superior antioxidant activity (EC_{50} 21.7 ± 1.2 μ M). However, unlike **9**, it was inactive against hBChE, suggesting that the 7-position is critical for enzyme affinity despite the 2-isomer resembling KYNA structure more precisely. Based on conducted bioassays **9** as MTDD targets four relevant aspects of AD e.i. neurotransmitter imbalance, OS and **9** as a privileged structure for binding metals prevents ion dyshomeostasis. Metal cation imbalance affects not only normal function of numerous enzymes and receptors, but also it is linked to A β accumulations, hyperphosphorylation of tau protein, and OS.

The compound, **10** (Fig. 15),⁶⁴ is KYNA modified by derivatization **A** and **B**. In this manner, a pharmacophore was created that combines use-dependent antagonism of voltage-sensitive sodium channels (VSNACs) and competitive antagonism at the strychnine-insensitive glycine site of NMDA receptors. VSNACs are essential for normal brain signaling, however, under pathological conditions their overactivation can cause excitotoxicity in early AD and as the disease progresses their overexposure to A β plaques can reduce their numbers which leads to

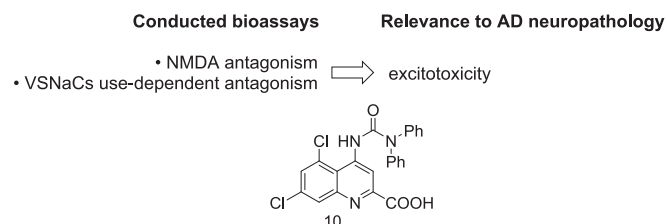


Fig. 15. The lead compound **10** from.64

impaired signal propagation.⁶⁵ Although **10** was assessed in a limited range of bioassays, largely restricting its proven scope to excitotoxicity and neuronal loss, the capacity of **10** to preferentially inhibit VGSCs that are firing at high frequencies (pathological state), while sparing those firing at normal physiological rates (use-dependent antagonism), confers a distinct therapeutic advantage over classical NMDA/mGluR5 antagonists.

Derivative **11** (Fig. 16) [66,67](#) is KYNA amide that incorporates N^1, N^1 -dimethylethane-1,2-diamine. It was synthesized to improve BBB penetration and subsequently tested using acute hippocampal slices from mouse brains to establish its ability preventing NMDA-induced excitotoxicity. The assay found **11** to be similarly neuroprotective as unmodified KYNA in young and adult mice, however, in old mice treatment with **11** treatment was not statistically significant compared to KYNA in the LDH (lactate dehydrogenase) release assay. [66](#) The same KYNA derivative was used in a study of drug release from bovine serum albumin (BSA) nanoparticle encapsulation which aimed to greatly improve the BBB penetration. [67](#) The drug release assay showed that only about 47% of the encapsulated **11** were released through a semi-permeable membrane compared to 71% of the pure **11**. This makes BSA encapsulation a great potential carrier for KYNA derivatives, unfortunately, the BBB permeability was measured only for unmodified KYNA with and without encapsulation. KYNA without colloidal carriers achieved a flux value slightly above 0.5×10^{-6} mol/s.cm², the permeability (Pe) was 4×10^{-7} cm/s. KYNA with BSA colloidal carrier (prepared at pH 7.40) had a flux of 3.7×10^{-6} mol/s.cm², the Pe was 16.3×10^{-7} cm/s. Overall, BSA nanoparticles achieved a three- to four-fold higher BBB permeability for KYNA compared to the unformulated drug. This increased permeability was attributed to enhanced membrane retention and the solubilizing effect of the BSA nanoparticles. Bioassays conducted in this study advocate compound **11** as MTDD with focus on two aspects of AD, specifically excitotoxicity/neuronal loss and neuroprotection.

2.6. Critical appraisal of KYNA-based MTDDs

Generally, conducting a direct comparison of bioassay results across compounds **1–11** is challenging due to the distinct methodologies employed in different studies. Even the most common Ellman's assay,⁶⁸ used in multiple sources^{51,57,58,60} for the determination of cholinesterase inhibitory activity varies in the source of the enzyme, substrate concentration, incubation time, and other parameters. Despite this inherent heterogeneity among bioassays [Table 1](#) summarizes the results from all relevant tests conducted with compounds **1–11**. With these nuances in mind, the rank order of potency regarding cholinesterase inhibition is: **9** (0.011 μM for hBuChE, **9** is inactive for hAChE) < **8** (0.046 μM for hAChE) < **7** (4.5 μM for hAChE) < **1** (11 μM for AChE) < **2, 3** (inactive for hAChE).

For other comparable bioassays, we prioritized *in vivo* models that capture the full physiological environment to ensure translational validity, rather than relying solely on *in vitro* screens. These other comparative bioassays, reported in multiple sources, include NMDA antagonism,^{51,64,66} ADME profiling models and antioxidant

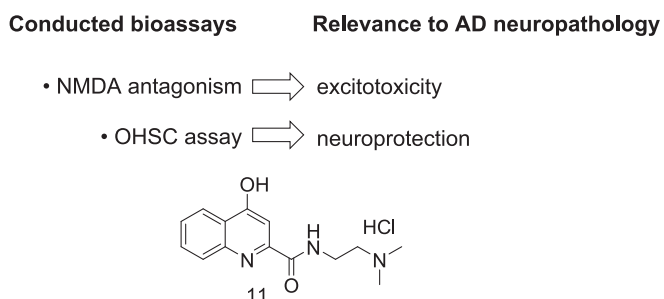
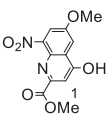
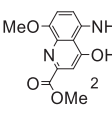
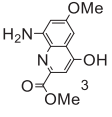
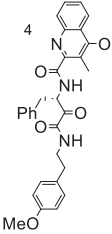
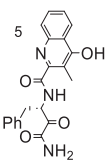
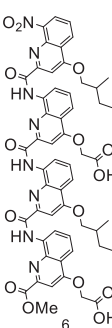
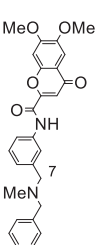
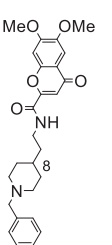


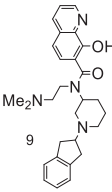
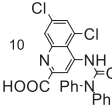
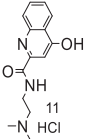
Fig. 16. The lead compound **11** from.[66,67](#)

Table 1
Comparative summary of bioassays conducted for compounds 8–18.

Structure	Primary Targets for inhibition (enzyme: IC ₅₀ ^a , Ki values etc.)	Secondary Bioassays (Antioxidant, Neurogenesis, etc.)	ADME & BBB Predictions (Method used)	Translational potential
	AChE: 11 μM; NMDA antagonism: 28% at 10 μM; mGluR5 antagonism: 44% at 10 μM	antioxidant (DPPH): Low	BBB: Low (possible hydrolysis during MDR1-MDCKII)	Compound tested <i>in vitro</i>
	AChE: >100 μM; NMDA antagonism: 16% at 10 μM; mGluR5 antagonism: –38% at 10 μM	antioxidant (DPPH): 1017 M ^{–1} s ^{–1} ; Aβ aggregation reduction: 57%	BBB: high 31 × 10 ^{–6} cm/s (MDR1-MDCKII); non-P-gp substrate	<i>In Vivo</i> protection: <i>C. elegans</i> (GMC101): reduced paralysis
	AChE: 11% at 10 μM; NMDA antagonism: 43% at 10 μM with Ki = 8 μM; mGluR5 antagonism: 68% at 10 μM with Ki = 0.8 μM	antioxidant (DPPH): 181 M ^{–1} s ^{–1}	BBB: high 29 × 10 ^{–6} cm/s (MDR1-MDCKII); non-P-gp substrate	Compound tested <i>in vitro</i>
	human μ-calpain: 0.71 μM	Not reported	Not reported	Compound tested <i>in vitro</i>
	human μ-calpain: 0.73 μM	Not reported	Not reported	Compound tested <i>in vitro</i>
	potent inhibitor of Aβ ₄₂ aggregation (complementary results from ThT fluorescence, MTS assay in N2a cells, ELISA, Dot Blot, TEM)	ROS scavenging: active (measured in <i>C. elegans</i> using the fluorescent dye H2DCFDA)	BBB: high (brain/plasma ratio in mice: 44%); metabolic stability: >90% (in human liver microsomes for 24 h)	<i>In Vivo</i> protection: <i>C. elegans</i> (GMC101 and CL2355): reduced paralysis and chemotaxis restored
	hAChE: 4.5 μM; hBuChE: >10 μM; hBACE-1: 6.7 μM; hLOX-5: 30.4 μM; σ1R: Ki = 0.53 μM; σ2R: Ki = 1.3 μM; hMAO-A/B: >100 μM	antioxidant (ORAC): inactive; neurogenesis: active (promotes NSC differentiation)	BBB: high 18.8 × 10 ^{–6} cm/s (PAMPA method)	Compound tested <i>in vitro</i>
	hAChE: 46 nM; hBuChE: >10 μM; hLOX-5: 74.3 μM; σ1R: Ki = 37.4 nM; σ2R: Ki = 0.24 μM; hMAO-A: 15.3 μM; hMAO-B: 5.2 μM	antioxidant (ORAC): inactive; neurogenesis: active (promotes NSC differentiation)	BBB: high 5.9 × 10 ^{–6} cm/s (PAMPA method)	Compound tested <i>in vitro</i>

(continued on next page)

Table 1 (continued)

Structure	Primary Targets for inhibition (enzyme: IC ₅₀ ^a , Ki values etc.)	Secondary Bioassays (Antioxidant, Neurogenesis, etc.)	ADME & BBB Predictions (Method used)	Translational potential
	hAChE: < 20% at 10 μM; hBuChE: 11 nM; metal chelation: active; redox cycle inhibition: active (ascorbate assay)	antioxidant (DPPH): EC ₅₀ = 61.4 μM; Intracellular ROS reduction: significant reduction (H ₂ O ₂ induced)	BBB: high logPe = -4.93 (approx. 11.7 × 10 ⁻⁶ cm/s) (PAMPA); non-P-gp substrate (Caco-2 with efflux ratio = 1.5)	Compound tested <i>in vitro</i>
	NMDA antagonism (glycine site): active; VSNAC use-dependent antagonism: active	Not reported	Not reported	Compound tested <i>in vitro</i>
	NMDA antagonism (glycine site): Active	neuroprotection: active (prevents LDH release in OHSC assay)	BBB: high, enhanced by hydrophilicity and BSA carrier (PAMPA method)	Ex Vivo (mouse slices): significantly reduced NMDA-induced neuronal loss

^a If the concentration of activity is not specified in the table it is IC₅₀.

properties.^{51,55,57,58,60,67}

The NMDA antagonism was assessed *via* three different methodologies. For compound **11**, an *ex vivo* (mouse slices) functional excitotoxicity assay was used. This method most closely mimics *in vivo* conditions and is therefore considered the most robust. However, it does not demonstrate direct receptor binding, only the neuroprotective outcome resulting from reduced NMDA-induced neuronal loss. For compound **10**, electrophysiology (TEVC) using *Xenopus* oocytes expressing NR1a/NR2A subunits was employed. Although this is an *in vitro* assay, it confirms that the drug prevents channel opening. Crucially, it can distinguish between competitive antagonists and channel blockers. Therefore, this method is more precise than the radioligand binding assays used for compounds **1–3**. This third assay is also *in vitro*, and it only establishes whether the tested compound binds to the receptor or not.

ADME profiling models are especially important as their liabilities are one of the primary barriers to AD drug development for KYNA derivatives. The reported MTDDs candidates have been tested predominantly for BBB penetration (**1–3**, **6–9**, **11**) and to a lesser extent for P-gp efflux (**2**, **3**, **9**). To establish whether a reported value of BBB penetration is sufficient we used the industry standards >4 × 10⁻⁶ cm/s for PAMPA-BBB,⁶⁹ >10 × 10⁻⁶ cm/s for MDR1-MDCKII,⁷⁰ and > 30% for *in vivo* brain/plasma ratio.⁷¹ Derivative **6** is the only compound assessed by *in vivo* brain/plasma ratio, making this the most reliable result. Derivatives **1–3** were assessed by MDR1-MDCKII which is an *in vitro* assay. However, it is a cell-based model, and it offers more physiological conditions than passive diffusion across an artificial lipid membrane (PAMPA method) used for compounds **7–9**, **11**. Despite unmodified KYNA having low BBB penetration all the tested KYNA derivatives achieved high BBB penetration. Furthermore, compounds **2**, **3**, **9** are not substrate for P-gp which is desirable against multi-drug resistance. This low efflux liability was established by two methods. The Caco-2 assay was used for **9** and offers higher physiological relevance as it utilizes human colon cancer cells. On the other hand, MDR1-MDCKII utilizes modified canine kidney cells, and was used for compounds **1–3**.

Antioxidant properties were determined by three bioassays. ROS scavenging for compound **6** was measured using an *in vivo* method, specifically measured in *C. elegans* using the fluorescent dye H2DCFDA. This method offers the greatest translational potential of the three. Compounds **7** and **8** were assessed by ORAC which is an *in vitro* assay that mimics physiological conditions to some extent. This method uses peroxyl radicals, which are the damaging radicals found in the human

body, and it is also performed in water. The least physiological accurate method is DPPH used for compounds **1–3**, **9**. All tested KYNA derivatives possess antioxidant properties (**2**, **3**, **6**, **9**) except **1**, **7**, **8**.

In summary, based on Table 1 we can identify two groups of highly promising MTDD candidates. Compounds **2** and **6** emerge as proof-of-concept leads, as they successfully showcased their efficacy and low toxicity *in vivo* (*C. elegans*). Specifically, they are both effective against the upstream inhibition of Aβ aggregation, they reduce ROS, reduce paralysis, compound **6** also restores chemotaxis, both effectively cross the BBB, and **2** is not a P-gp substrate. Their main disadvantage is low inhibitory activity against AChE, and low antagonism of NMDA and mGluR5 receptors. From MTDD potential they hit three out of six relevant contributors to AD. Specifically they target Aβ plaques, oxidative stress and they are neuroprotective, however they are ineffective or were not tested for neurotransmitter imbalance, tau tangles, or neuroinflammation. Compounds **7** and **8** stand out for their high potency. They both highly penetrate the BBB, and they hit five out of six AD contributors. Specifically, they are both effective against neurotransmitter imbalance (inhibition of hAChE), oxidative stress (only compound **8**) (hMAO-A/B), neuroinflammation (LOX-5), Aβ aggregation (only compound **7**) (upstream inhibition of hBACE-1). Furthermore, they are not only neuroprotective, but they also stimulate neurogenesis (stimulation of σ1R, σ2R and NSC differentiation assay) which offers a unique disease-modifying potential. The only AD contributor that they were not tested against is hyperphosphorylated tau protein. Their disadvantage is that all the bioassays were conducted *in vitro*, which lowers their transitional value.

Additionally, certain empirical observations can be made about modification to the KYNA structure to achieve improvement of properties and an MTDD character. The most recurring strategy is amidation (derivatization C) as it allows for targeted alteration of final attributes in one step by utilizing bioactive amine. This approach possibly expands the bioactivity of the MTDD and creates a pro-drug strategy. The amidation showcased the improvement of BBB penetration in several discussed compounds. The amide bond can possibly be cleaved by enzymes in the brain after the MTDD crosses the BBB. For example, compound **7** could release modified KYNA with neuroprotective and anti-inflammation properties with experimental drug AP2238 (amine moiety) with AChE and BuChE inhibitory properties. However, the actual mechanism of action in human body would most likely be more complex than this simple example.

Derivatization A is the second most common approach, especially

with multiple small electron-donating moieties. Lastly, derivatization **B** is used infrequently for specific tasks, as is the case in foldamer **6** and VSNaCs antagonist **10**.

The MTDD design of the presented compounds **1–11** usually integrates the established neuroprotective and anti-inflammatory features of KYNA with neurotransmitter modulation, anti-A β aggregation activity, and, less frequently, with antioxidant properties and the maintenance of ion homeostasis.

2.7. Future perspectives for KYNA-based MTDDs

The evolution of KYNA derivatives into viable MTDDs requires overcoming pharmacokinetic hurdles (BBB penetration, low solubility) and slow receptor dissociation kinetics for NMDA. Both properties are intrinsic limitations of the unmodified KYNA scaffold. Enhancement of BBB penetration can be achieved by direct synthetic modification of KYNA to mask the polar hydroxy (enol tautomer) or amino (oxo tautomer) and carboxylic acid groups. This is the predominant option in current literature, especially *via* amidation with a bioactive amine. After the amide crosses the BBB, it could be enzymatically cleaved thus this option could be also labeled as a carrier-linked prodrug strategy. An alternative pro-drug strategy that could be explored is the use of a bioprecursor to cross the BBB, specifically modified kynurenine.⁷² It is also possible to leverage colloidal platforms like BSA nanoparticles. This option was studied only on compound **11**, however, it can achieve three- to four-fold increase in BBB permeability.⁶⁷

While NMDA antagonism was established for multiple discussed compounds (**1–3,10–11**) none of the studies reported kinetic rates of dissociation, leaving this as a possible target for future research.

Another possible future research focus could be the integration of anti-tau pharmacophores into the MTDD profile of KYNA derivatives. There were no direct bioassays to establish the efficacy of compounds **1–11**, in this regard, only indirect links, such as μ -calpain inhibition (**4–5**) and metal chelation (**9**), which were linked to tau hyperphosphorylation. The addition of anti-tau properties could enable KYNA MTDDs to address all the main aspects of AD.

In the future ML / AI could be employed to elucidate many SARs connected not only to bioactivity, but also to pharmacokinetic issues. However, this approach currently has its limitations,⁷³ specifically limited data sets (it would require substantially more KYNA derivatives and bioassays to train the model). Creating these quality studies should be one of the priorities in the field.

3. Conclusion

The presented review underlines the growing need for novel treatments for Alzheimer's disease, especially with a multi-target directing drug character due to the complex pathology of the disease. From the conducted literature search, it is clear that kynurenic acid has a beneficial impact on slowing down Alzheimer's neurodegeneration, though the precise mechanism of action is yet to be determined. Possible mechanistic candidates are neuroprotective and anti-inflammatory attributes of kynurenic acid. The necessity to better understand role of kynurenic acid in Alzheimer's disease is evident from the rising number of recently published reviews written on this topic. However, to fully utilize the potential of kynurenic acid as a drug, it is necessary to overcome certain unfavorable characteristics such as low solubility, inability to cross the blood-brain barrier, and the very slow reversibility of its antagonism to the *N*-methyl-D-aspartate receptor. These hurdles can be surmounted by rational molecular design. At the same time the development of kynurenic acid derivatives can be employed to achieve multi-target directing drug character, especially by amidation (derivatization **C**) with bioactive amines. This approach expands on bioactivity of MTDD and facilitates possible a carrier-linked pro-drug strategy to enhance pharmacokinetic profile. The field of kynurenic acid derivatives is still in its infancy, therefore, this review aims to provide guidance and

accelerate the research in this area by summarizing the synthesis of kynurenic acid, which is currently dominated by the Conrad–Limpach synthesis. While this synthesis has its drawbacks (thermal cyclisation step) it already exists in modifications accounting for green chemistry protocols and flow chemistry that partially circumvent its limitations. Furthermore, the review discusses the most frequent modifications to kynurenic acid, the most promising kynurenic acid derivatives **1–11**, and the results from their conducted bioassays. Eleven leading structures were identified in the literature that, as potential multi-target directing drugs, integrate the established properties of kynurenic acid with modulation of neurotransmitters, inhibition of amyloid-beta aggregation, and the prevention of oxidative stress and neuroinflammation. Among these, compounds **2** and **6** emerged as “translational winners” with proven *in vivo* efficacy (*C. elegans*), while compounds **7** and **8** stood out as “potency winners,” exhibiting the broadest multi-target profiles that include the unique ability to stimulate neurogenesis. Future iterations of these derivatives should incorporate kinetic rates of dissociation study for NMDA antagonism and aim to integrate anti-tau pharmacophores in MTDD profiles of KYNA derivatives.

Declaration of generative AI in scientific writing

During the preparation of this work the author Branislav Pavilek used Gemini LLM in order to minimize grammatical and stylistic errors, especially to avoid overuse of the same words. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

CRediT authorship contribution statement

Branislav Pavilek: Writing – original draft. **Dana Hajduová:** Conceptualization. **Dušan Bortňák:** Writing – review & editing. **Viktor Milata:** Supervision, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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